Pearson Edexcel International A Level · Pure Mathematics 3

Chapter 5 — Exponentials and logarithms Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P3 第5章 · 指数与对数
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P3-C05-R01 · P3 · Chapter 5

For $x > 1$, $\ln(x+1) - \ln(x-1) = \ln 3$.

(a) Find x exactly. (3)

(b) Find the exact value of $\ln(x^2 - 1)$ at this solution. (2)

(Total for Question 1 is 5 marks)

A cooling model is $T = 18 + 62e^{-0.2t}$, where $t \geq 0$ is in minutes and T is in °C.

(a) Find the initial temperature and limiting temperature. (2)

(b) Find the time when $T = 30$ °C, giving an exact expression and a value to 2 decimal places. (4)

(Total for Question 2 is 6 marks)

3.

$e^{2x} - 5e^{x} + 4 = 0.$

(a) Find all exact solutions. (4)

(b) Find all real solutions of $e^{2x} - 5e^{x} + 4 < 0$. (3)

(Total for Question 3 is 7 marks)

Two quantities are related by $y = Ax^n$, with $A > 0, x > 0$. A plot of $\ln y$ against $\ln x$ is a straight line through $(0, \ln 3)$ and $(\ln 2, \ln 24)$.

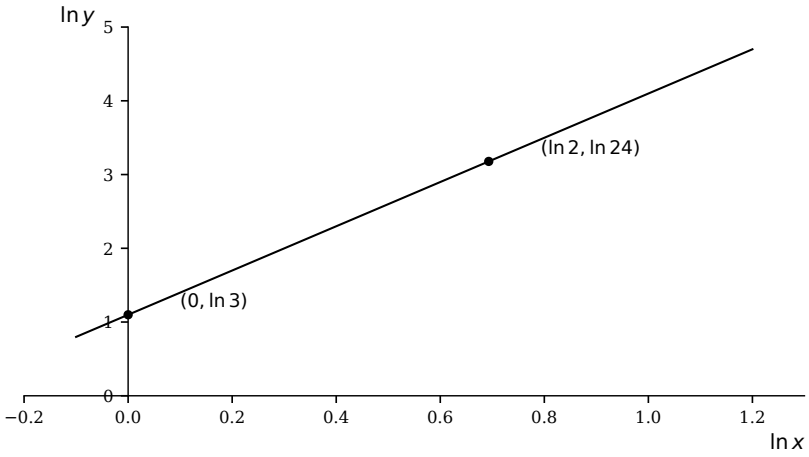


Figure 4

(a) Find A, n. (4)

(b) Find x when $y = 81$. (2)

(Total for Question 4 is 6 marks)

5.

A quantity satisfies $M = Ae^{kt}$. It has values 12 at $t = 1$ and 96 at $t = 4$.

(a) Find exact A, k. (4)

(b) Find the time when $M = 150$, to 3 decimal places. (3)

(Total for Question 5 is 7 marks)

6. P3-C05-R06 · P3 · Chapter 5

The curve is $y = \ln(4 - x)$.

(a) State its domain, vertical asymptote and axis intersections. (3)

(b) Find all x for which $y > \ln 2$. (3)

(Total for Question 6 is 6 marks)

$\ln x + \ln(x - 2) = \ln 8$, where $x > 2$.

(a) Solve exactly. (4)

(b) Find the exact value of $\ln x - \ln(x - 2)$ at the solution. (2)

(Total for Question 7 is 6 marks)

8. P3-C05-R08 · P3 · Chapter 5

A material has mass $m = 60e^{-kt}$, where $k > 0$. After 8 days its mass is 45.

(a) Find k exactly. (3)

(b) Find its half-life and the time to reach one quarter of its initial mass, in exact form. (3)

(Total for Question 8 is 6 marks)

9. P3-C05-R09 · P3 · Chapter 5

For $x > 0$, $y = Be^{ax}$. A graph of $\ln y$ against x has gradient $-3/2$ and vertical intercept 2.

(a) Find a, B. (3)

(b) Find the exact positive x at which $y = 1$. (2)

(Total for Question 9 is 5 marks)

$2e^{2x} - 7e^{x} - 4 = 0$.

(a) Find all exact real solutions, showing why one substituted root is inadmissible. (4)

(b) Find the value of e^{-x} at the solution. (2)

(Total for Question 10 is 6 marks)

The function is $f(x) = \ln(x^2 + 1)$.

(a) State its domain, range and every axis intersection. (4)

(b) Solve $f(x) = \ln 5$ and $f(x) < \ln 5$ separately. (3)

(Total for Question 11 is 7 marks)

A growth model is $P = 200e^{kt}$, with t in years. The population doubles every 3 years.

(a) Find k exactly. (2)

(b) For this model, find

(i) the population after 9 years, (2)

(ii) the least whole number of years at which the population is at least 1000. (3)

(Total for Question 12 is 7 marks)

$\ln(x+3) = 2\ln x$, where $x > 0$.

(a) Find the exact solution. (4)

(b) Find the exact value of $\ln((x+3)/x)$ at the solution. (2)

(Total for Question 13 is 6 marks)

The curve is $y = Ae^{2x} + B$. Its horizontal asymptote is $y = -3$, and it passes through $(\ln 2, 13)$.

(a) Find A, B. (3)

(b) Find its exact x-intercept and y-intercept. (3)

(Total for Question 14 is 6 marks)

For $x > 1$, $F(x) = \ln\left(\dfrac{x+1}{x-1}\right)$.

(a) Show that $F(x) > 0$. (3)

(b) Find the input at which $F(x) = \ln 5$. (3)

(Total for Question 15 is 6 marks)

A signal follows $S = 12e^{-0.4t}$ and a background signal follows $B = 3e^{-0.1t}$, for $t \geq 0$.

(a) Find exactly when $S = B$. (3)

(b) Find the interval of time when $S > B$. (3)

(Total for Question 16 is 6 marks)

A relation $y = Ae^{kx}$ has positive observations $(0, 5)$ and $(2, 20)$.

(a) Find A, k exactly. (3)

(b) Find $\ln y$ at $x = -1$ and give the corresponding exact y. (3)

(Total for Question 17 is 6 marks)

$\ln x = \ln(2 - x) + \ln 3$.

(a) State the domain restrictions and solve. (4)

(b) Find $\ln[x(2 - x)]$ at the solution. (2)

(Total for Question 18 is 6 marks)

The resistance of a component is $R = R_0 e^{kT}$. It is 10 ohms at $T = 0$ and 15 ohms at $T = 20$.

(a) Find R_0, k exactly. (4)

(b) Find the temperature change that multiplies resistance by $3/2$, and explain why it is independent of the starting temperature. (3)

(Total for Question 19 is 7 marks)

The graph of $Y = \ln(y - 2)$ against x is a straight line with gradient $-1/2$ and intercept $\ln 6$.

(a) Find y in terms of x. (3)

(b) Find the exact x at which $y = 5$ and state the horizontal asymptote of the original graph. (3)

(Total for Question 20 is 6 marks)

$e^x + e^{-x} = 5/2$.

(a) Find all exact solutions. (4)

(b) Find all possible values of $e^x - e^{-x}$ at these solutions. (3)

(Total for Question 21 is 7 marks)

A quantity is modelled by $Q = 100(1 - e^{-kt})$, where $k > 0$, $t \geq 0$. At $t = 4$, $Q = 75$.

(a) Find k exactly and the limiting value of Q. (4)

(b) Find the time at which $Q = 93.75$. (3)

(Total for Question 22 is 7 marks)

Let $f(x) = \ln(3x - 2)$.

(a) Find f^{-1} with its domain and range. (4)

(b) Solve $f(x) = 2f(1)$. (3)

(Total for Question 23 is 7 marks)

For $x > 0$, $(\ln x)^2 - \ln x - 6 = 0$.

(a) Find all exact solutions. (4)

(b) Find all x for which $(\ln x)^2 - \ln x - 6 \leq 0$. (3)

(Total for Question 24 is 7 marks)

The curves $y = e^{2x}$ and $y = 9e^{-x}$ meet.

(a) Find the intersection exactly. (3)

(b) Find the horizontal distance between the intersection and the point on $y = e^{2x}$ where $y = 1$. (3)

(Total for Question 25 is 6 marks)

A power-law model is $y = Ax^n$. A base-10 log plot has intercept 2 and gradient -3.

(a) Find A, n and the value of y at $x = 2$. (4)

(b) State the gradient and intercept if natural logarithms are used instead on both axes. (2)

(Total for Question 26 is 6 marks)

27.

For real x, $F(x) = \ln(e^x + 1) - x$.

(a) Show that $F(x) = \ln(1 + e^{-x})$. (3)

(b) Find x when $F(x) = \ln 3$. (3)

(Total for Question 27 is 6 marks)

28. P3-C05-R28 · P3 · Chapter 5

A model for concentration is $C = 8e^{-0.3t} + 2$, with $t \geq 0$.

(a) Find the concentration after 5 time units, to 3 significant figures. (2)

(b) Find the time when the concentration first falls below 3, to 2 decimal places, and state the exact boundary. (4)

(Total for Question 28 is 6 marks)

For $x > 0$, $\ln(\sqrt{x}) + \ln(x^2) = 5$.

(a) Find x exactly. (3)

(b) Find the exact value of $x^{\ln x}$ at the solution. (3)

(Total for Question 29 is 6 marks)

Two measurements obey $y = Ae^{-kx} + B$, where $k > 0$. The horizontal asymptote is $y = 3$, and the values at $x = 0, 2$ are $11, 5$.

(a) Find exact A, B, k. (4)

(b) For the same model, find

(i) the value of y at $x = 4$, (2)

(ii) the exact x at which $y = 4$. (3)

(Total for Question 30 is 9 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P3-C05-R01 · 5 marks

(a) · 3 marks

$x = 2$.

合并商：(x+1)/(x−1)=3，解一次式，检查x>1。

M1	对数商法则
M1	线性求解
A1	合法根

(b) · 2 marks

$\ln 3$.

代x2得$x^2-1=3$，真数为正。

M1	正确代入
A1	值

Question 2

P3-C05-R02 · 6 marks

(a) · 2 marks

80 °C and 18 °C.

t0时指数1；t增大时衰减项趋0。

B1	初温
B1	极限温度

(b) · 4 marks

$t = 5\ln(31/6) \approx 8.21$ min.

62e^{−.2t}=12，取ln得−.2t=ln(6/31)。

M1	隔离指数
M1	自然对数
A1	精确时间
A1	精度

Question 3

P3-C05-R03 · 7 marks

(a) · 4 marks

$x = 0, \ln 4.$

u=e^x>0，(u−1)(u−4)=0，取ln回解。

M1	正变量代换
M1	因式分解
M1	还原
A1	两根

(b) · 3 marks

$0 < x < \ln 4.$

二次式在1

M1	二次不等式
M1	单调映回x
A1	区间

Question 4

P3-C05-R04 · 6 marks

(a) · 4 marks

$A = 3, n = 3.$

lny=nlnx+lnA，截距ln3；斜率ln(24/3)/ln2=3。

M1	正确线性化
M1	截距求A
M1	斜率求n
A1	参数

(b) · 2 marks

$x = 3.$

$81=3x^3$，x正给3。

M1	代回原模型
A1	x

Question 5

P3-C05-R05 · 7 marks

(a) · 4 marks

$k = \ln 2, A = 6.$

相除e^{3k}=8给kln2，再由Ae^k12求A6。

M1	两条件相除
M1	对数求k
M1	回求A
A1	精确参数

(b) · 3 marks

$t = \ln 25 / \ln 2 \approx 4.644.$

6e^{tln2}=150，tln2=ln25。

M1	正确阈值方程
M1	对数
A1	时间

Question 6

P3-C05-R06 · 6 marks

(a) · 3 marks

$x < 4$; $x = 4$; $(3, 0)$ and $(0, \ln 4)$.

真数4−x正；y0给4−x1；x0给ln4。

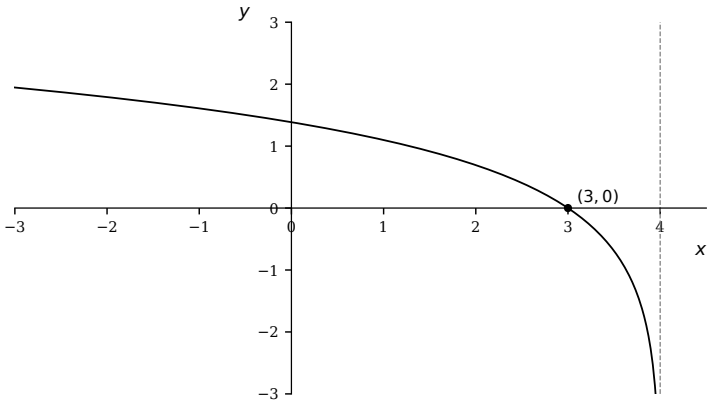
B1	域与渐近线
M1	横截距
A1	两点

(b) · 3 marks

$x < 2$.

ln严格递增，4−x>2，结果已包含在原域中。

M1	合法对数比较
M1	原域检查
A1	范围



Suggested sketch

Question 7

P3-C05-R07 · 6 marks

(a) · 4 marks

$x = 4$.

x(x−2)=8，方程(x−4)(x+2)=0，只留4。

M1	合并对数转乘积
M1	二次求根
M1	原域筛选
A1	x

(b) · 2 marks

$\ln 2$.

ln[4/2]=ln2，不能相减真数后直接取ln。

M1	对数商法则
A1	值

Question 8

P3-C05-R08 · 6 marks

(a) · 3 marks

$k = \frac{1}{8}\ln(4/3)$.

e^{-8k}=3/4，ln后求k。

M1	正确比值
M1	自然对数
A1	k

(b) · 3 marks

Half-life $8\ln 2/\ln(4/3)$; quarter time $16\ln 2/\ln(4/3)$ days.

e^{-kt}=1/2或1/4，对数指数分别ln2与2ln2，所以时间加倍。

M1	半衰关系
M1	四分之一关系
A1	两时间

Question 9

P3-C05-R09 · 5 marks

(a) · 3 marks

$a=-3/2, B=e^{2}.$

lny=ax+lnB，直读斜率和对数截距，再指数还原B。

M1	正确线性形式
A1	a
A1	B

(b) · 2 marks

$x=4/3.$

lny=0给−3x/2+2=0。

M1	使用对数直线求零值
A1	x

Question 10

P3-C05-R10 · 6 marks

(a) · 4 marks

$x = \ln 4$; reject $e^x = -1/2$.

(2u+1)(u−4)=0，u=e^x严格正。

M1	指数代换分解
M1	排负u
M1	ln还原
A1	唯一根

(b) · 2 marks

$1/4$.

e^{-x}=1/e^x=1/4。

M1	指数倒数关系
A1	值

Question 11

P3-C05-R11 · 7 marks

(a) · 4 marks

Domain $\mathbb{R}$; range $[0, \infty)$; only intercept $(0, 0)$.

$x^2+1 \geqslant 1$，ln输出$\geqslant 0$，在x0取0，任意更大真数可取。

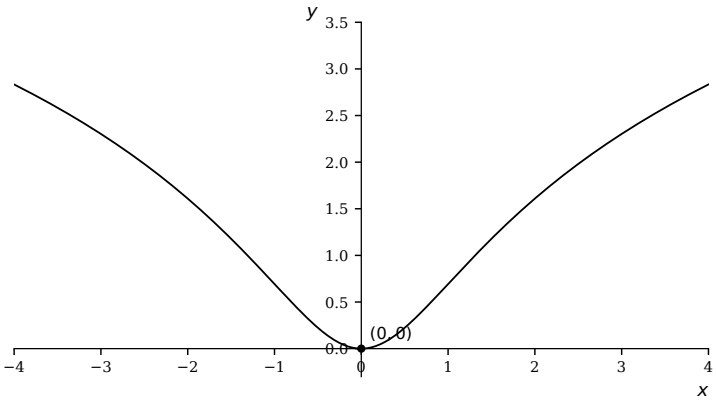
M1	真数恒正
M1	最小输出
A1	域与范围
B1	截距

(b) · 3 marks

Equality $x = \pm 2$; inequality $-2 < x < 2$.

ln单调，分别化为$x^2+1=5$或<5。

M1	合法去对数
M1	平方方程与不等式
A1	两个解集



Suggested sketch

Question 12

P3-C05-R12 · 7 marks

(a) · 2 marks

$k = \ln 2/3.$

e^{3k}=2，取ln。

M1	倍增条件
A1	k

(b) For this model, find

(i) · 2 marks

1600.

9年等于3个倍增周期，所以200×2^{3}。

M1	周期倍数
A1	数值

(ii) · 3 marks

7 years.

$t \geqslant 3\ln 5/\ln 2 \approx 6.966$，最小整数7，等号条件为至少。

M1	正确阈值不等式
M1	整数取上界
A1	年数

Question 13

P3-C05-R13 · 6 marks

(a) · 4 marks

$x = (1 + \sqrt{13})/2.$

$\ln(x+3)=\ln x^2$给$x^2-x-3=0$，正根唯一满足$x>0$。

M1	对数幂法则
M1	二次公式
M1	正域筛选
A1	根

(b) · 2 marks

$\ln x.$

由$x+3=x^2$，商为x，真数正。

M1	使用根关系化简商
A1	值

Question 14

P3-C05-R14 · 6 marks

(a) · 3 marks

$A = 4, B = -3$.

渐近线给B，e^{2ln2}=4，13=4A−3。

B1	B
M1	精确指数还原
A1	A

(b) · 3 marks

x-intercept $(\frac{1}{2}\ln(3/4), 0)$; y-intercept $(0, 1)$.

令4e^{2x}=3取ln；x0时4−3=1。

M1	指数零值方程
M1	对数求x
A1	两截距

Question 15

P3-C05-R15 · 6 marks

(a) · 3 marks

$(x+1)/(x-1) = 1 + 2/(x-1) > 1$, so its natural logarithm is positive.

利用x>1使分母正，真数严格大于1，ln单调且ln1=0。

M1	真数改写
M1	正值比较
A1	严格结论

(b) · 3 marks

$x = 3/2$.

(x+1)/(x−1)=5给x+1=5x−5，x1.5合法。

M1	去对数并清合法分母
M1	求解检查
A1	x

Question 16

P3-C05-R16 · 6 marks

(a) · 3 marks

$t = \frac{10}{3}\ln 4$.

相除4e^{-0.3t}=1，对数后t=ln4/.3。

M1	相比隔离指数
M1	ln运算
A1	精确时间

(b) · 3 marks

$0 \le t < \frac{10}{3}\ln 4$.

4e^{-0.3t}>1，随t严格减，所以在首次等值之前；包括初始时点。

M1	正确不等式方向
M1	原时间域
A1	区间

Question 17

P3-C05-R17 · 6 marks

(a) · 3 marks

$A = 5, k = \ln 2.$

首点A5，e^{2k}=4所以kln2。

B1	A
M1	对数求k
A1	k

(b) · 3 marks

$\ln y = \ln(5/2);\ y = 5/2.$

用lny=ln5+xln2，在−1处相减；也可代指数模型。

M1	正确对数线性值
M1	还原
A1	两值

Question 18

P3-C05-R18 · 6 marks

(a) · 4 marks

$0 < x < 2$; solution $x = 3/2$.

两真数均正；合并成ln[3(2−x)]，得到x=6−3x。

M1	两真数正域
M1	合并对数
M1	解并查域
A1	解

(b) · 2 marks

$\ln(3/4)$.

代1.5×.5=3/4。

M1	正确乘积代值
A1	对数值

Question 19

P3-C05-R19 · 7 marks

(a) · 4 marks

$R_0 = 10$, $k = \ln(3/2)/20$.

首条件读R0，第二e^{20k}=3/2后取ln。

B1	R0
M1	观测比例
M1	对数
A1	k

(b) · 3 marks

20 degrees; $R(T+\Delta)/R(T) = e^{k\Delta}$.

相除时初值和e^{kT}抵消，固定比值只由温差决定。

M1	正确比值式
M1	消去起始温度
A1	温差和理由

Question 20

P3-C05-R20 · 6 marks

(a) · 3 marks

$y = 2 + 6e^{-x/2}$.

ln(y−2)=−x/2+ln6，指数还原再加2。

M1	写线性式
M1	指数还原
A1	模型

(b) · 3 marks

$x = 2\ln 2$; asymptote $y = 2$.

6e^{-x/2}=3给e^{-x/2}=1/2，求x；指数衰减极限加回2。

M1	对数求x
A1	精确x
B1	原坐标渐近线

Question 21

P3-C05-R21 · 7 marks

(a) · 4 marks

$x = \pm \ln 2$.

设u=e^x>0，u+1/u=5/2，2u²−5u+2=0给u2或1/2。

M1	正变量代换
M1	二次因式
M1	ln还原
A1	两x

(b) · 3 marks

$\pm 3/2$.

可分别代u，或平方差关系：(sum)²−(difference)²=4，差平方9/4。

M1	代值或平方恒等式
M1	两符号
A1	值

Question 22

P3-C05-R22 · 7 marks

(a) · 4 marks

$k = \ln 4/4$; limit 100.

1−e^{-4k}=.75给e^{-4k}=.25；t大时指数趋0。

M1	校准方程
M1	对数
A1	k
B1	极限

(b) · 3 marks

$t = 8$.

此时指数=1/16，为t4处1/4的平方，所以时间8；或用ln求。

M1	正确指数比例
M1	求时间
A1	精确值

Question 23

P3-C05-R23 · 7 marks

(a) · 4 marks

$f^{-1}(x) = (e^x + 2)/3$, domain $\mathbb{R}$, range $(2/3, \infty)$.

原域x>2/3，原值域全实；反解e^y=3x−2，交换域范围。

M1	正确指数反解
A1	反函数
B1	逆域
B1	逆范围

(b) · 3 marks

$x = 1$.

f1=ln1=0，所以右边0，3x−2=1。

M1	先求f1
M1	反解零值
A1	x

Question 24

P3-C05-R24 · 7 marks

(a) · 4 marks

$x = e^{3}, e^{-2}$.

令u=lnx，(u−3)(u+2)=0；两实u都可指数还原为正x。

M1	对数代换
M1	因式分解
M1	指数回代
A1	两x

(b) · 3 marks

$e^{-2} \leq x \leq e^{3}$.

二次条件−2⩽lnx⩽3，指数单调，闭端点保留。

M1	二次范围
M1	单调指数化
A1	闭区间

Question 25

P3-C05-R25 · 6 marks

(a) · 3 marks

$(\frac{2}{3}\ln 3, 3^{4/3})$.

e^{3x}=9给x=ln9/3，纵坐标e^{2x}=9^{2/3}。

M1	合并指数
M1	ln求x
A1	精确交点

(b) · 3 marks

$\frac{2}{3}\ln 3$.

第二点x0，第一点横坐标正，差即该值。

M1	找零对数位置x0
M1	横差
A1	距离

Question 26

P3-C05-R26 · 6 marks

(a) · 4 marks

$A = 100, n = -3$; $y = 25/2$.

logy=logA+nlogx，截距2意味着A100而不是2；代x2。

M1	对数截距还原
B1	n
M1	代原模型
A1	y

(b) · 2 marks

Gradient -3; intercept $\ln 100 = 2\ln 10$.

lny=nlnx+lnA，换底不改变幂指数斜率，但截距依底而变。

M1	自然对数线性式
A1	两参数

Question 27

P3-C05-R27 · 6 marks

(a) · 3 marks

$x = \ln(e^x)$, so combine as a quotient $(e^x + 1)/e^x = 1 + e^{-x}$.

分母e^x严格正，日志商法则始终合法。

M1	将x改写为lne^x
M1	商化简
A1	目标式

(b) · 3 marks

$x = -\ln 2$.

1+e^{-x}=3给e^{-x}=2，求x。

M1	使用化简式
M1	指数解
A1	精确根

Question 28

P3-C05-R28 · 6 marks

(a) · 2 marks

3.79.

C5=8e^{-1.5}+2≈3.78504。

M1	正确代值
A1	精度

(b) · 4 marks

$t > \frac{10}{3}\ln 8 \approx 6.93$; boundary $(10/3)\ln 8$.

8e^{−.3t}<1，−.3t<−ln8，除负数方向翻转。

M1	正确严格指数条件
M1	不等号翻转
A1	精确边界
A1	数值

Question 29

P3-C05-R29 · 6 marks

(a) · 3 marks

$x = e^2$.

左边(1/2+2)lnx=(5/2)lnx，故lnx2。

M1	对数幂法则
M1	合并系数
A1	x

(b) · 3 marks

e^4.

在解处lnx2，因此x^{lnx}=(e²)²=e⁴。

M1	使用已得对数值
M1	指数法则
A1	精确值

Question 30

P3-C05-R30 · 9 marks

(a) · 4 marks

$B = 3, A = 8, k = \ln 2.$

基线3，初值A8，5=8e^{-2k}+3给e^{-2k}=1/4。

B1	B
M1	初值求A
M1	对数求k
A1	参数

(b) For the same model, find

(i) · 2 marks

$7/2.$

e^{-4ln2}=1/16，8/16+3=3.5。

M1	正确指数缩放
A1	y

(ii) · 3 marks

$x = 3.$

8×2^{-x}=1给2^x8，x3。

M1	正确基线扣除
M1	指数求解
A1	x